

Intuition Engine Programmer's Reference Manual

Contents

Part I - Intuition Engine BASIC

1. BASIC Programming Rules
2. BASIC Language Vocabulary

Part II - Programming Graphics

3. Display Model Overview
4. VideoChip
5. VGA Text and Graphics Modes
6. TED Video
7. ANTIC and GTIA
8. ULA Display
9. Voodoo 3D Rasterizer
10. Tile and Sprite Layers from BASIC

Part III - Programming Sound and Music

11. Audio Architecture Overview
12. SoundChip and SFX
13. PSG and AY-3-8910
14. SN76489
15. SID Family
16. TED Audio
17. POKEY and SAP Playback
18. AHX Engine
19. MOD Playback
20. WAV Sample Player
21. Paula DMA Engine
22. Music from BASIC and from each CPU

Part IV - BASIC to Machine Language

23. Memory Model and MMIO Map
24. IE64
25. IE32
26. 6502
27. Z80
28. M68K
29. x86
30. Processor Timing, Traps, and Exceptions

- 31. Coprocessor and Cross-CPU Calls
- 32. IE Mon - the Machine Monitor
- 33. IE Script

Part V - Input / Output Guide

- 34. Disk and File I/O
- 35. The HOST Command
- 36. Keyboard, Mouse, Controller MMIO
- 37. Serial Devices

Appendices

A. EhBASIC Keyword Abbreviations and Token Map B. Screen and Character Codes C. ASCII and CHR\$ Tables D. Per-Engine MMIO Maps E. Music Note and Frequency Tables F. Math and Derivative Helpers G. Per-CPU Opcode Quick Reference H. Per-CPU Symbol Index I. Error Message Index J. Full Memory Map K. Block Diagrams